Compressive Hyperspectral Target Detection With Restricted Distribution Property

Qingyao Yang, Xiaoqin Wang, Dingyi Wang, Baoying Yu, Yumei Zhou, and Shushan Qiao, *Member, IEEE*

***Abstract*— Compressive sensing (CS) is a widely utilized technique in hyperspectral imaging front end, and hyperspectral target detection (HTD) in the CS domain is crucial for conserving computational and storage resources on front-end platforms and achieving real-time detection. However, existing HTD methods can only handle reconstructed hyperspectral images. To address this limitation, this study investigates a novel property under hyperspectral CS and proposes an algorithm framework for HTD without reconstruction. The newly proposed property, named restricted distribution property (RDP), is based on assumptions about spectral random vectors and derivations from the CS model. It indicates that the spectral vectors in the compressed hyperspectral follow a deterministic Gaussian distribution under specific conditions and provide a generalized expression for their covariance matrix. Building upon this property, a Kullback–Leibler (KL) divergence subspace distribution estimator is developed, enabling HTD without reconstruction. Experimental result demonstrates that the performance of the proposed detector in compressed hyperspectral data is comparable to or even superior to conventional methods in original hyperspectral images. The result is inspiring for future research.**

***Index Terms*— Compressive sensing (CS), hyperspectral, restricted distribution property (RDP), target detect.**

NOMENCLATURE

CS	Compressive sensing.
HTD	Hyperspectral target detection.
RIP	Restricted isometry property.
REP	Restricted entropy property.
RSP	Restricted spectrum property.
RDP	Restricted distribution property.

I. INTRODUCTION

HYPERSPECTRAL technology, an advanced imaging technology, holds significant promise with extensive applications. Its exceptional spectral resolution enables precise material detection in hyperspectral target detection (HTD) [1], [2], widely used in military reconnaissance [3], [4], environmental monitoring [5], [6], [7], and mineral exploration [8], [9], [10].

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Traditional HTD methods, such as orthogonal subspace projection (OSP) [11], adaptive coherence estimator (ACE) [12], and constrained energy minimization (CEM) algorithm [13], rely on spectral matching for background and target separation. CEM, a notable algorithm, applies constraints using FIR filters to minimize responses and achieve effective HTD. Various enhancements have evolved from CEM, including h-CEM cascaded multilayered CEM detectors [14] and ensemble-based CEM (E-CEM) [15]. Hyperspectral representations are also widely used in HTD. For example, the combinatorial sparse and collaborative representation (CSCR) detector uses target and background dictionaries for effective target detection [16]. The sparse-based target detection (STD) detector reconstructs hyperspectral images through a spare dictionary and extracts targets from the reconstruction residuals [17]. Recent explorations in HTD include deep learning (DL) methods. For example, HTD-Net trains autoencoders to learn discriminative similarity by pixel pairs [18]. BLTSC combines CEM for initial target detection with a constrained adversarial autoencoder (AAE) for background refinement, achieving target separation [19]. HTD-IRN [20] utilizes a lightweight multiscale transformer to learn the subspace representations of both background and target. MLSN [21] utilizes deep residual convolution for spectral vector embedding into Euclidean space through metalearning.

All these HTD methods above belong to conventional back-end processes, in which hyperspectral data are transmitted to a processing platform for preprocessing followed by HTD. Fig. 1 illustrates the difference between traditional back-end and snapshot front-end HTD processing frameworks. Conversely, onboard processing on the front end like satellites and UAVs [23], [24] enables real-time detection [23], [25], [26], [27], [28], providing immediate feedback for tasks such as military reconnaissance and target tracking [29]. Front-end HTD also reduces onboard storage requirements and transmission bandwidth, significantly easing the burden on resource-constrained platforms [27].

Conventional methods cannot be directly applied to hyperspectral before reconstruction, which prevents them from the front-end HTD, especially in the snapshot front end with CS. They, based on the prior target spectrum, search for the candidate target with similar spectral features. However, the spectrums and features in CS are completely different from the original domain. Moreover, the CS sensing matrix and sparse matrix may differ or be unknown for different platforms, making it more challenging to identify specific spectral features for detection. Although hyperspectral CS data can be

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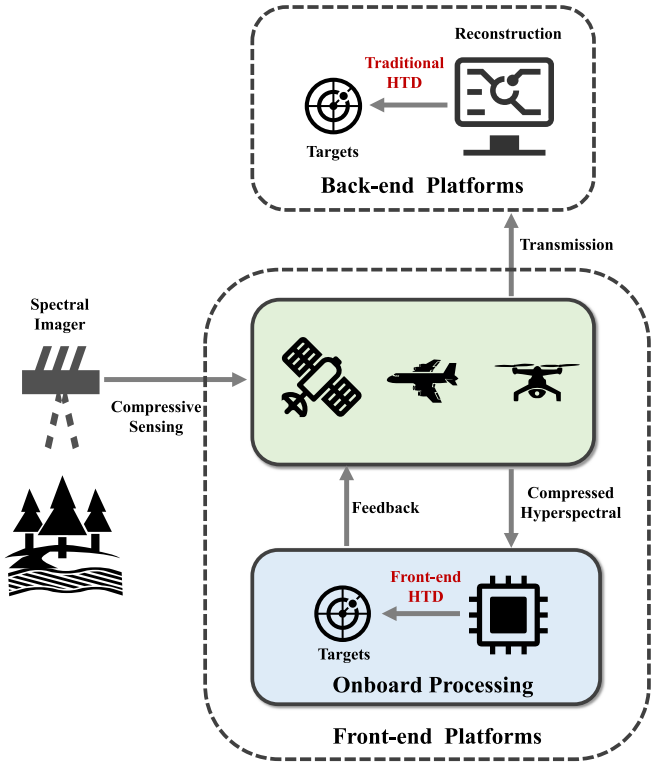


Fig. 1. Distinction between traditional and compressive HTDs in the front-end platform.

reconstructed by solving nonlinear optimization [30], [31], the time consumption is unacceptable for real-time detection in front-end HTD.

On the other hand, conventional HTD algorithms often struggle with the impact of spectral variability [32]. The spectrum collected from the same object may differ due to atmospheric compensation, geometric distortions, and sensor noise [33]. A single-target spectrum used in the experiment may not fully represent the spectrum in real sensor output, thus affecting detection accuracy.

To solve the problems above, this article explores the distribution of spectral random vectors undergoing CS and develops a brand new restricted distribution property (RDP). We consider that the target spectrum in the original domain, due to various disturbances, exhibits spectral variability, constituting a range of random vectors with an unknown probability density function. In a situation of a universal sensed model [34], we analysis the redistribution of the random vector through the linear transformation by sensing matrix in CS, finding that each element of the spectral vector in CS follows a Gaussian distribution related to the variance in the original domain. RDP ensures that the probability distribution characteristics of spectral random vectors are preserved during the CS process. It extends properties of the spectrum (such as RIP, RSP, and REP [35]) to probabilistic attributes, adapting spectral variability. Algorithms developed based on this property, once deployed on platforms satisfying the universal sensed model, are unaffected by spectral variability, variations in the sensing matrix, and sparse transformation matrix.

Based on RDP, this article introduces a method of compressive HTD without reconstruction. The prior vectors are seen as the samples of spectral random vectors, deriving the target distribution in CS through RDP. Then, a neighborhood subspace in CS is extracted to estimate the distribution of the center spectrum. Using Kullback–Leibler (KL) divergence as a similarity measurement, the proposed method searches for spectrums with a distribution similar to the target as detection results.

This article introduces a novel RDP with an analysis of spectral random vectors' redistribution in CS while proposing a front-end HTD method without reconstruction. The contributions of this article are given as follows.

1) Developing a novel RDP in CS, which preserves the probabilistic characteristics of spectral random vectors, the covariance of special random vectors is derived based on RDP.
2) Proposing a subspace sliding windows detector based on RDP to achieve compressive HTD, with KL divergence as an estimation of spectral similarity; Introducing a uniform band shuffle strategy among subspaces, and enhancing the overall information entropy and detection performance.
3) Conducting experiments to verify RDP, as well as comparison experiments on four compressed datasets with different compression ratios (CRs) to demonstrate the effectiveness of the proposed method.

II. Relative Works

A. Universal Sensed Model

Over the past few decades, CS has gradually emerged as a crucial technology in the fields of information theory, communication, and signal processing. CS is a sensing paradigm that, under appropriate conditions, enables the recovery of the signal of interest using a strategy at sub-Nyquist sampling rates. This process differs from traditional analog-to-digital conversion, as it directly acquires compressed information and can reduce sampling speeds and data storage. The CS comprises two key processes: sparse sampling of the signal and recovery, allowing the signal to be obtained in a compressed form and subsequently recovered using reconstruction algorithms.

A hyperspectral image $Y \in \mathbb{R}^{H\times W\times L}$ can be decomposed into spatial dimensions of size $H \times W$ and a spectral dimension of length L. The process of sparsely sampling the spectral vectors from the original L spectral bands to M spectral bands as a linear combination, where $M \ll L$, can be expressed as

$$y = \Phi\Psi r + n \tag{1}$$

where $y \in \mathbb{R}^M$ represents the sparsely sampled spectral vector, $r \in \mathbb{R}^L$ is the spectral vector in the original domain, $\phi \in \mathbb{R}^{M\times L}$ is the sensing matrix, $\psi \in \mathbb{R}^{L\times L}$ is the sparse matrix, and n denotes noise.

In the CS theory, it is required that the original signal r exhibits sparsity, or it can be transformed into a sparse signal using a sparse transform matrix ψ. Meanwhile, the sensing matrix ϕ should be as incoherent as possible with the sparse

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matrix ψ to ensure a better reconstruction of signal [36]. The coherence is defined as (2), where ϕ_j and ψ_j represent the columns of ϕ and ψ

$$1 \le \mu(\phi, \psi) = \sqrt{L} \quad \max_{1 \le j,i \le L} |\langle \phi_j, \psi_i \rangle| \le \sqrt{L}. \tag{2}$$

The number of compressed bands is restricted by a lower bound (3), where c is a specified constant and k is the order of sparsity

$$M \ge ck \log (L/k). \tag{3}$$

Besides, it is also important for the sensing matrix to satisfy the RIP to guarantee that relative measures of distance are maintained and preserved between the original and compressed signal domains [37]. The RIP can be expressed as the following inequality:

$$(1 - \delta_k)\|r\|_2^2 \le \|\Phi r\|_2^2 \le (1 + \delta_k)\|r\|_2^2 \tag{4}$$

where δ_k is the restricted isometry constant in the range of 0 to 1.

The significance of RIP can be understood as follows: for any k-sparse vectors, after transforming by the sensing matrix ϕ, their Euclidean distances remain unchanged. Random sensing matrices have been previously proven to satisfy the RIP and are incoherent with any sparse matrix [37].

On that basis, Della Porta et al. [34] propose a universal sensed model by constructing a Gaussian distributed sensing matrix ϕ_G, where each element ϕ_{ij} satisfies $\phi_{ij} \sim \mathcal{N}(0, 1/L)$. In this case, the sparse matrix ψ does not need to be known during signal acquisition, only during signal reconstruction. CS under the universal sensed model can be simplified to the following formula:

$$y = \Phi_G r + n. \tag{5}$$

This indicates that, under the universal sensed model, the algorithm can proceed without the need to precisely know the sparse matrix. Moreover, all the information is retained in the CS signal, as confirmed in [37].

Based on the universal sensed model, recent research [35] further develops the REP and RSP in CS. Similar to how RIP ensures the Euclidean distance preservation of spectral vectors, REP indicates the preservation of entropy for CS spectral vectors. RSP means that the similarity between two spectral vectors measured by spectral information divergence (SID) can be retained under CS, resembling that restricted conformal property (RCP) [38] preserves the angle between two spectral vectors measured by spectral angle mapping (SAM).

B. KL Divergence

KL divergence, as a nonparametric measure of similarity, holds significance in information theory and finds extensive applications across various domains. The general formula for computing KL divergence is expressed as in (6), where t and q represent two probability distributions

$$\mathrm{KL}(t \mid q) = \int t(x) \log \frac{t(x)}{q(x)} \, \mathrm{d}x. \tag{6}$$

This can also be commonly denoted as

$$\mathrm{KL}(t \mid q) = -\int t(x) \log q(x) \mathrm{d}x + \int t(x) \log t(x) \mathrm{d}x. \tag{7}$$

In the case where both distributions t and q follow nonmultivariate Gaussian distributions, based on prior research [39], the expression can be further simplified, as shown in (8). This formula indicates that for nonparametric similarity between two nonmultivariate Gaussian distributions, it is possible to calculate using the mean and variance of their sample, avoiding integration of the complex probability density functions. In the next section, this article will demonstrate that spectral vectors under CS satisfy the conditions for utilizing the simplified KL divergence formula

$$\begin{aligned} \mathrm{KL}(t \mid q) &= -\int t(x) \log q(x) \mathrm{d}x + \int t(x) \log t(x) \mathrm{d}x \\ &= \frac{1}{2} \log(2\pi\sigma_q^2) + \frac{\sigma_t^2 + (\mu_t - \mu_q)^2}{2\sigma_q^2} \\ &\quad - \frac{1}{2}(1 + \log(2\pi\sigma_t^2)) \\ &= \log \frac{\sigma_q}{\sigma_t} + \frac{\sigma_t^2 + (\mu_t - \mu_q)^2}{2\sigma_q^2}. \end{aligned} \tag{8}$$

III. Restricted Distribution Property

A. Derivation of RDP

In the universal sensed model mentioned above, the spectral vectors are sparsely sampled from the original L bands to M bands through CS of hyperspectral bands. In this model, ϕ_G is a Gaussian random sensing matrix, where each element satisfies $\phi_{ij} \sim \mathcal{N}(0, (1/M))$. Here, r represents the original signal, and y represents the sparsely sampled signal. Utilizing (5), the compressed spectral vector y_t of the target r_t can be expanded as follows:

$$y_t = \begin{pmatrix} \phi_{11} & \cdots & \phi_{1L} \\ \cdots & \cdots & \cdots \\ \phi_{M1} & \cdots & \phi_{ML} \end{pmatrix} \begin{pmatrix} r_t(1) \\ \cdots \\ r_t(L) \end{pmatrix}. \tag{9}$$

In CS, each element of the spectral vector is the inner product of the row of the matrix with the original spectrum r_t, expressed as

$$y_t(i) = \sum_{l=1}^{L} \phi_{il} r_t(l). \tag{10}$$

Denote each element as $x_t(l) = \phi_{il} r_t(l)$, and then,

$$y_t(i) = \sum_{i=1}^{L} x_t(l). \tag{11}$$

Due to factors such as spectral variability, the spectral vector r_t in the original domain is not a deterministic vector but a random vector. Each element $r_t(l)$ on this random vector follows an unknown distribution. By obtaining prior information on the target spectral vector, although the exact mathematical model of its distribution may not be known, the expected value $\mu_t(l)$ and variance $\sigma_t^2(l)$ of each spectral vector's discrete point can be obtained.

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Therefore, the expectation of each term $x_t(l)$ can be computed as follows:

$$\begin{aligned} E[x_t(l)] &= E(\phi_{il})E(r_t(l)) \\ &= 0 \end{aligned} \tag{12}$$

as well as the variance

$$\begin{aligned} \mathrm{Var}[x_t(i)] &= \mathrm{Var}[\phi_{il} r_t(l)] \\ &= \left(\mu_\phi * \sigma_t(l)\right)^2 + \left(\mu_t(l) * \sigma_\phi\right)^2 + \sigma_t^2(l)\sigma_\phi^2 \\ &= \frac{1}{M}\mu_t^2(l) + \frac{1}{M}\sigma_t^2(l). \end{aligned} \tag{13}$$

Denote $B_L^2 = \sum_{l=1}^{L} \mathrm{Var}[x_t(l)]$, constructing the discriminant formula for the Lyapunov central limit theorem

$$\lim_{L\to\infty} \frac{1}{B_L^{\delta+2}} \sum_{l=1}^{L} [E|x_t(l) - E[x_t(l)]|] = 0. \tag{14}$$

Due to $E[x_t(l)] = 0$, which implies that in (14), $E[x_t(l) - E[x_t(l)]] = 0$, there exists $\delta > 0$ make the Lyapunov discriminant formula equal 0. Therefore, it can be considered that when the number of original spectral bands L is large, $y_t(i) = \sum_{l=1}^{L} x_t(l)$ satisfies the Lyapunov central limit theorem. In other words, the standardized $y_t(i)$ follows a standard normal distribution. Consequently, it can be derived that the distribution of $y_t(i)$ satisfies

$$\frac{\sum_{l=1}^{L} x_t(l) - \sum_{l=1}^{L} 0}{B_L^2} \sim \mathcal{N}(0, 1) \tag{15}$$

$$y_t(i) = \sum_{l=1}^{L} x_t(l) \sim \mathcal{N}\left(0, B_L^2\right). \tag{16}$$

From expression (16), it can be concluded that although CS in the universal sensed model may lose the per-element information of the original spectral random vector, the mean and variance of its distribution are preserved in B_L^2. Moreover, despite the unknown distribution function of the original spectral random vector r_t, the elements of the spectral vector under CS still follow a Gaussian distribution. We refer to this property as the RDP, with B_L^2 being the RDP coefficient.

Compared with other CS properties that consider spectral vectors as random vectors, REP preserves the entropy for a spectral vector and RIP preserves the vector length [35], RDP establishes the correlation between the distribution of the random vector in the original domain and CS. With RDP, we can accurately estimate the distribution function of the spectral vector in CS by sampling the original spectral random vector.

The Gaussian random sensing matrix ϕ_G under the universal sensed model ensures that each element of the spectral vector in CS satisfies the Lyapunov central limit theorem. This is a crucial condition for satisfying RDP, similar to the role played by the Gaussian random sensing matrix in RIP, REP, and RSP [35], [37].

B. Covariance With RDP

The covariance matrix is an effective means of describing statistical information and variations between hyperspectral bands. In [40], preservation of sample covariance is derived, which indicates that the covariance matrix C_ϕ under CS approaches the original domain covariance C as the number of compressed bands M approaches the original band number L. This section investigates the covariance matrix under the premise of RDP.

In assumption, the target in the original domain is a stationary random vector r_t with a constant mean vector μ_t. The spectral vector of the target under CS can be represented as $y_t = \phi r_t$. The covariance matrix C_ϕ of y_t is expressed by (17). Simplify the formula using $E[y_t] = 0$ based on RDP

$$\begin{aligned} C_\phi &= E\left[\left(y_t - E\left[y_t\right]\right)^T \left(y_t - E\left[y_t\right]\right)\right] \\ &= E\left[y_t^T y_t\right]. \end{aligned} \tag{17}$$

According to the CS formula, the expression of C_ϕ can be replaced by ϕr_t

$$\begin{aligned} C_\phi &= E\left[(\phi r_t)^T \phi r_t\right] \\ &= E\left[r_t^T \phi^T \phi r_t\right]. \end{aligned} \tag{18}$$

Using $\phi^T\phi = I + e$, C_ϕ can be expressed as

$$\begin{aligned} C_\phi &= E\left[r_t^T (I + e) r_t\right] \\ &= E\left[r_t^T r_t\right] + E\left[r_t^T e r_t\right]. \end{aligned} \tag{19}$$

As $\phi^T\phi \to I$ when the dimension $M \to L$ in the CS [35], the error matrix e decreases, and it can be considered that the covariance matrix C_ϕ of y_t under CS tends to $E[r_t^T r_t]$.

Next, the covariance matrix C of the random vector r_t in the original domain can be expressed by the following formula:

$$\begin{aligned} C &= E\left[(r_t - \mu_t)^T (r_t - \mu_t)\right] \\ &= E\left[\left(r_t^T - \mu_t^T\right)(r_t - \mu_t)\right] \\ &= E\left[r_t^T r_t - r_t^T \mu_t - \mu_t^T + \mu_t^T \mu_t\right]. \end{aligned} \tag{20}$$

Assuming that the target spectrum is a stationary random vector with a constant mean vector μ_t, the covariance matrix C can be expressed as

$$C = E\left[r_t^T r_t\right] - E\left[r_t^T\right]\mu_t - \mu_t^T E[r_t] + \mu_t^T \mu_t. \tag{21}$$

With $E[r_t^T] = \mu_t^T$ and $E[r_t] = \mu_t$, further simplifying C as

$$C = E\left[r_t^T r_t\right] - \mu_t^T \mu_t. \tag{22}$$

In the general case, the covariance matrix of the random vector in the original domain satisfies (22). When the dimension $M \to L$ under CS, the covariance matrix $C_\phi \to E[r_t^T r_t]$. In addition, the absence of a linear relationship between the first-order moments between the bands in the original domain, i.e., satisfying $\mu_t^T \mu_t = 0$, can result in $C_\phi \to C = E[r_t^T r_t]$. Therefore, the preservation of sample covariance can be seen as a special case.

IV. Detection Method

In this section, we describe the details of the proposed HTD method based on RDP. The framework is illustrated in Fig. 2.

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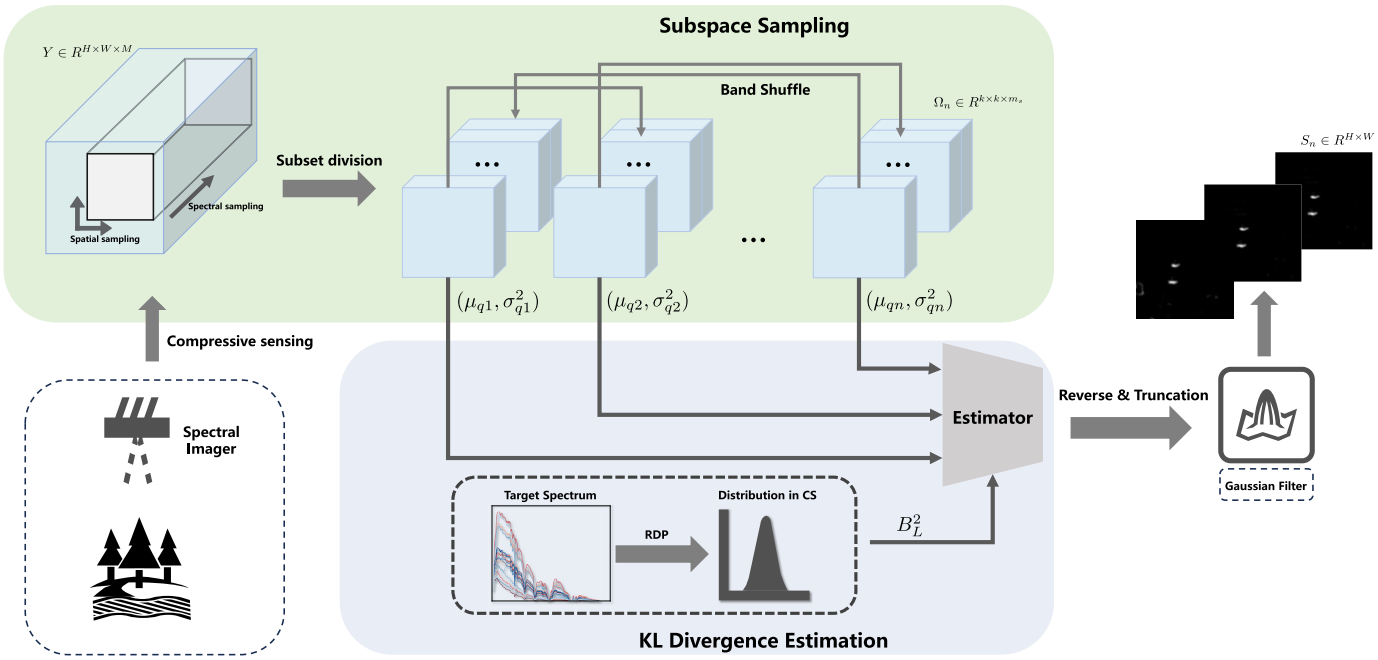


Fig. 2. Framework of the proposed method.

A. Subspace Sampling

RDP is a description of the distribution of each element in the spectral vector. The key to spectral target detection based on RDP lies in how to effectively estimate the distribution of compressed hyperspectral samples. This section proposes a sampling method for neighborhood subspaces, sampling simultaneously in spatial and spectral dimensions.

There are two bases for this sampling method. First, according to (15), elements in the same spectral vector located at different wavelengths follow the same distribution in CS. Sampling in the spectral dimension essentially involves selecting samples from multiple observations that follow the same distribution. Second, in the spatial direction, the small neighboring region of the pixel to be detected can usually be considered to have the same material composition as the pixel or, due to factors such as spectral mixing, have spectra similar to that of the pixel. Therefore, sampling in the spatial dimension can also be seen as observations of the distribution of the central pixel.

Let the spatial coordinates of y_q in the global hyperspectral image be (a, b), and the global 3-D hyperspectral image is $Y \in \mathbb{R}^{H\times W\times M}$; then, the spectral vector can be expressed as $y_q = Y(a, b, :)$.

Then, extract the spectral vectors in the $k \times k$ neighborhood of y_q as a sliding window and sampling subspaces of size $k \times k \times M$, where M is the number of bands in CS. Divide the extracted subspace into several subsets Ω_n each of size $k \times k \times m_s$, where m_s is the number of bands in each subset, and the total number of subsets is N. The subspace $\Omega_n \in \mathbb{R}^{k\times k\times m_s}$ can be represented as

$$\Omega_n = \{Y(a+i, b+j, m) \mid i, j, m \in \mathbb{Z}\} \tag{23}$$

where $((-k-1)/2) \le i, j \le ((k-1)/2)$, $(n-1)m_s + 1 \le m \le nm_s$, and $N \times m_s = M$.

From an information theory perspective, increasing the entropy within subsets helps enrich internal details, thereby improving detection effectiveness. Enhancing the entropy within subsets can be achieved by reducing the correlation between internal bands [42] and increasing the randomness of the arrangement of internal bands. Therefore, random band shuffle is performed to exchange the bands between the divided subsets

$$\text{Bandshuffle}(\Omega_1, \Omega_2, \ldots, \Omega_N). \tag{24}$$

It is worth noting that the proposed RDP is adapted to the compressed 2-D hyperspectral image, such as DD-CASSI [41]. Due to the introduction of the dual disperser, which counteracts the spatial mixing between bands caused by dispersion, satisfying (14), which restricts only the number of original bands.

The subspace sampling approach is supposed to be adjusted as the compressed 2-D image has only one band in the spectral dimension, and its sample distribution can only be obtained from the neighborhood of a 2-D plane, rather than a 3-D subspace. Hence, setting the number of subsets N to 1 and m_s to 1, removing band shuffling, and slightly increasing the neighborhood size k can enable adaptation to this compressed 2-D image.

B. KL Divergence Estimation

The RDP from the above section implies that the distribution of the target spectral vector under CS is $y_t(i) \sim \mathcal{N}(0, B_L^2)$. Denote the prior target spectral vector as $r_t = [r_{t1}, r_{t2}, \ldots, r_{td}]^T$, in which d is defined as the number of target signature samples and $r_{td} \in \mathbb{R}^{1\times L}$. Through (11) and (13), the RDP coefficient can be calculated

$$B_L^2 = \frac{1}{M}\sum_{l=1}^{L}\left(\mu_t^2(l) + \sigma_t^2(l)\right) \tag{25}$$

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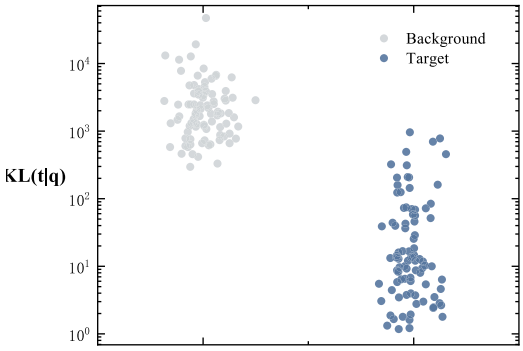


Fig. 3. KL divergence of target and background.

where $\mu_t(l)$ and $\sigma_t^2(l)$ are the mean and variance of the element at position l obtained from this series of sample vectors

Assuming the distribution of the spectrum to be detected in CS is $y_q(i) \sim \mathcal{N}(\mu_q, \sigma_q^2)$, both are nonmultivariate Gaussian variables. Therefore, the simplified KL divergence from (8) can be used to measure the similarity between the two distributions as a criterion for detecting the target.

Let t and q denote the probability density functions of the prior target and the detected spectrum in CS, respectively. Substituting these along with the RDP coefficients into (8), we get

$$\mathrm{KL}(t \mid q) = \log \frac{\sigma_a}{B_L} + \frac{B_L^2 + \mu_q^2}{2\sigma_q^2}. \tag{26}$$

According to (26), it is necessary to know the sample mean μ_{qn} and variance σ_{qn}^2 of the detected subsets Ω_n to calculate the KL divergence

$$\mu_{qn} \equiv \frac{1}{k^2 m_s} \sum_{i,j,m} Y(a+i, b+j, m) \tag{27}$$

$$\sigma_{qn}^2 \equiv \frac{1}{k^2 m_s} \sum_{i,j,m} (Y(a+i, b+j, m) - \mu_{qn})^2. \tag{28}$$

Substitute them into (26) to calculate the KL divergence as the detection result $S_n(a, b)$ for this spectral vector in subset n

$$S_n(a, b) = \log \frac{\sigma_{qn}}{B_L} + \frac{B_L^2 + \mu_{qn}^2}{2\sigma_{qn}^2}. \tag{29}$$

Equation (29) measures the similarity between two distributions. The KL divergence tends to be zero when they resemble; conversely, it sharply increases when the distributions are dissimilar. The background pixels result in a divergent value in S_n, as shown in Fig. 3; the background values are almost 10^1–10^2 times larger than the value of the target pixel.

As the increase in the discrepancy between the sampling distribution and the target spectral distribution from RDP, the KL divergence value of (29) grows exponentially. This leads to a significant contrast between the KL divergence values of background and target, enabling effective target segmentation in the KL divergence space.

The sharp increase in KL divergence values expands the dynamic range of the detection map dramatically, approximately ranging from 1 to 5 $\times$ 10^4, as shown in Fig. 3. Within this high dynamic range, only values close to 0 are relevant for detection results. Therefore, by inverting and truncating (30), the dynamic range is confined to the possible range of target presence, thereby excluding numerous pixels far from the target distribution

$$S_n(a, b) = \begin{cases} 0, & S_n(a, b) \geq \mathrm{TH} \\ \mathrm{TH} - S_n(a, b), & S_n(a, b) < \mathrm{TH}. \end{cases} \tag{30}$$

The Gaussian filter is applied to the results obtained from each subset to eliminate the background noise while smoothing the target. Finally, merge them as the output of the detection

$$D = \sqrt{\sum \text{Gauss-filter}[S_n]^2}. \tag{31}$$

The overall method is further described in Algorithm 1.

Algorithm 1 Compressive HTD

1. Initial condition:
Let $Y \in \mathbb{R}^{H \times W \times M}$ be the hyperspectral image in CS, M is dimensionality of compressed hyperspectral image.
Let $r_t = [r_{t1}, r_{t2}, \ldots, r_{td}]^T$ be the samples of target spectral random vectors in original domain. And $r_{td} \in \mathbb{R}^{1 \times 1 \times L}$, L is the number of bands in orginal domain, d is the number of target spectrum samples.
2. RDP process:
Get B_L^2 from the mean and variance of r_t.
3. HTD in sliding windows:
for$((a, b)$ in $(H, W))$:
(a). $\{\Omega_1, \Omega_2, \ldots, \Omega_n\}$ in center of $y_q = Y(a, b, :)$
(b). Bandshuffle$(\Omega_1, \Omega_2, \ldots \Omega_n)$
(c). $\mu_q = [\mu_{q1}, \mu_{q2}, \ldots \mu_{qn}]^T \leftarrow \Omega$
(d). $\sigma_q^2 = [\sigma_{q1}^2, \sigma_{q2}^2, \ldots \sigma_{qn}^2]^T \leftarrow \Omega$
(e). $[S_1(a, b), S_2(a, b), \ldots, S_n(a, b)]^T = \mathrm{KL}$ with μ_q, σ_q^2 and B_L^2
(f). **if** $S_n(a, b) \geq \mathrm{TH}$, $S_n(a, b) = 0$;
else $S_n(a, b) = \mathrm{TH} - S_n(a, b)$
result $=\| \text{Gauss-filter}[S_n] \|_2$

V. Experiment

A. Comparison Methods

In this experiment, the detection performance of the current algorithms is compared in original hyperspectral images with the detection performance of our method under CS. The selected comparative algorithms include ACE, ECEM, OSP, SMF, HTD-IRN [20], and MLSN. It is noteworthy that these algorithms are not directly applicable to HTD in CS, so the results are derived from reconstructed hyperspectral based on the OMP method [43].

In the experiments, the comparison algorithms were configured with optimal parameters. Specifically, for ECEM, the window size was set to [1/4, 2/4, 3/4, 4/4], the number of detection layers and CEM per layer were set to 10 and 6, respectively, and λ was set to 1e-6. Simultaneously, in the

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TABLE I
DETAILS OF THE DATASETS

	Sensor	captured place	resolution	Size	Original band number
Sandiego	AVIRIS	Sandiego Airport	3.5m	100*100	184
ABU-Airport	AVIRIS	Los Angeles	7.1m	100*100	205
ABU-Beach	AVIRIS	Bay Champagne	4.4m	100*100	188
ABU-Urban	AVIRIS	Texas Coast	17.2m	100*100	204

training of HTD-IRN, the iteration count was set to 5000, weight decay was set to 2e-5, learning rate (lr) was set to 1e-3, the number of background endmembers was set to 30, regularization parameter of sparse loss was set to 10, and ϵ was set to 4. In MLSN training, the margin, iteration count, and lr were set to 1, 50, and 1e-3, respectively.

In our proposed algorithm, results under different CRs are compared. Here, L represents the original number of spectral bands, m is the number of bands in CS, and CR is defined as m/L. A smaller CR indicates a higher compression level applied to the hyperspectral image.

B. Dataset

The datasets used in this experiment include the Sandiego dataset and the ABU dataset.

1) Sandiego Dataset: The first dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the San Diego airport area in California, USA. The images used in this experiment were not preprocessed to remove noise. The scene samples and ground truth are shown in the first and second columns of Fig. 5, respectively. Detailed information is provided in Table I.

2) ABU Dataset: Another three scenes of the Airport-Beach-Urban (ABU) dataset [44] were used and can be obtained from a public website. The first and second columns of Fig. 5 show scene samples and ground truth, respectively. The selected scenes include the airport, beach, and urban areas, all captured by the AVIRIS sensor. Detailed information is provided in Table I.

A Gaussian sensing matrix was constructed based on (1) to perform CS on all datasets, generating a series of compressed hyperspectral images for detection. To construct prior information for the targets, all target spectrums from the ground truth are extracted and used as samples for the spectral random vector. Mean and variance are calculated for each discrete point, and the target distribution under CS was constructed based on (13) and (16). The RDP coefficient B_L^2 was then computed.

C. Evaluation Criteria

The evaluation criteria include the receiver operating characteristic (ROC) analysis [45] and the area under the curve (AUC). The ROC is composed of the detection rate (P_d), false alarm rate (P_f), and threshold (τ). Three curves, namely, (P_d, P_f), (P_d, τ), and (P_f, τ), are formed by combining these components. AUC represents the area enclosed by the ROC curve and provides a more accurate reflection of the detector's performance compared to the ROC curve. The (P_d, P_f) curve serves as an indicator of the tradeoff between the detection rate and the false alarm rate, with a higher AUC indicating better detection performance. The (P_f, τ) curve is designed to assess the background suppression ability of the detector, where a higher or lower AUC reflects better background suppression capability.

In addition, two composite metrics of three AUCs for all-around estimation are added [46], which can be expressed as follows:

$$\mathrm{AUC}_{\mathrm{OA}} = \mathrm{AUC}(P_d, P_f) + \mathrm{AUC}(P_d, \tau) - \mathrm{AUC}(P_f, \tau) \tag{32}$$

$$\mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}(P_d, \tau)}{\mathrm{AUC}(P_f, \tau)}. \tag{33}$$

To evaluate the proposed RDP, quantile–quantile (QQ) plots are used as measurement [47]. QQ plots serve as a valuable tool in statistical analysis for assessing the goodness-of-fit between sample data and theoretical distribution. By comparing observed quantiles to those expected from a chosen distribution, QQ plots provide a visual indication of distributional conformity. A linear arrangement of points suggests a good fit, while deviations may signify departures from the theoretical distribution.

D. RDP Analysis

All target spectral vectors from the ABU-Airport dataset are extracted, and the RDP coefficient B_L^2 is calculated for the theoretical distribution $N(0, B_L^2)$. First, the target spectral vector $y \in \mathbb{R}^{1\times1\times M}$ in CS is obtained as the sampling distribution and plotted QQ plots for different CRs of $[0.1, 0.25, 0.5]$, as shown in Fig. 4(a). Then, to further verify if the subspace in its neighborhood also satisfies RDP, subspaces of size $\Omega \in \mathbb{R}^{k\times k\times m}$ are extracted from the target region in CS as the sampling distribution and plotted QQ plots. Fig. 4(b) and (c) depicts the results for subspaces with sizes $k = 3, m = [6, 10, 14]$ and $k = 5, m = [6, 10, 14]$, respectively, and both in CR $= 0.5$.

The results in Fig. 4 indicate that using the CS spectral vectors as samples, the points on the QQ plots are scattered along the line of the theoretical distribution $N(0, B_L^2)$. This suggests that the sampling can be well-fit by the theoretical distribution based on RDP. Moreover, when examining the QQ plots for subspace sampling, it is evident that the RDP theoretical distribution provides a good fit.

This observation confirms the existence of the RDP property in CS under a universal sensed model. The distribution of CS spectral vectors can be accurately computed based on the probability characteristics of the spectral random vectors in the original domain. Furthermore, the subspace sampling

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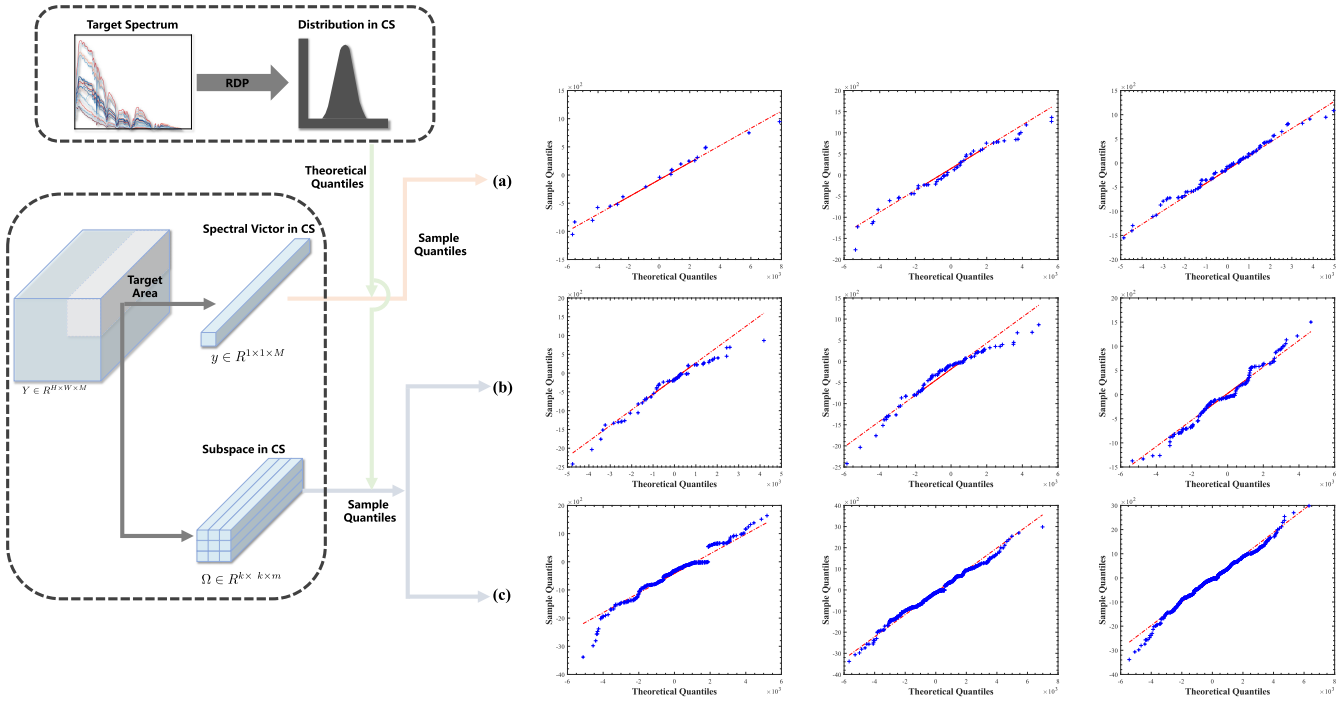


Fig. 4. QQ of RDP theoretical distribution and samples. (a) Spectral vector in CR = [0.1, 0.25, 0.5]. (b) Subspace in size of [3 × 3 × 6], [3 × 3 × 10], [3 × 3 × 14], and CR in 0.5. (c) Subspace in size of [5 × 5 × 6], [5 × 5 × 10], [5 × 5 × 14], and CR in 0.5.

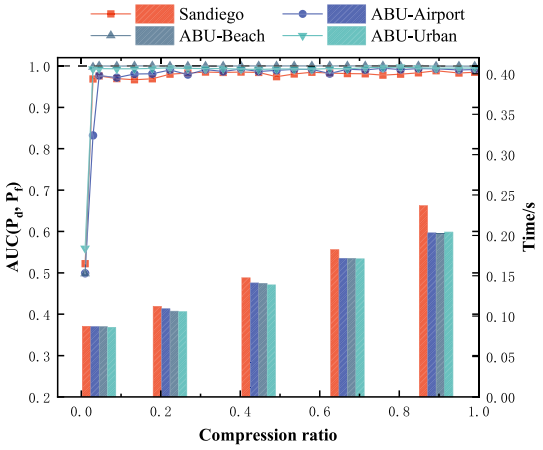


Fig. 5. AUC(P_d, P_f) and processing time in different CRs.

method mentioned in the previous section is also applicable to RDP. Moreover, as the number of sampled points within a small neighborhood increases, the fitting of the distribution improves. This ensures the accuracy of the algorithm developed based on subspace sampling.

E. HTD Results

1) Comparison of CRs: The AUC(P_d, P_f) values and processing time for different CRs between 0.01 and 0.99 are compared in Fig. 5. After CR increases to approximately 0.1, the proposed algorithm consistently maintains a high AUC(P_d, P_f). At a CR of 0.9, the processing time is around 0.2 s, which decreases as CR decreases.

2) Comparison of Detection Maps: The detection maps for various algorithms are presented in Figs. 6–9. Subplots (a) and (b) show the original image and ground truth, respectively. Subplots (c)–(f) display the detection results for the proposed algorithm at CR = [0.1, 0.25, 0.5, 0.75], while (g)–(l) show the results of ACE, ECEM, OSP, SMF, HTD-IRN, and MLSN in the original domain.

For the Sandiego dataset, ACE and HTD-IRN achieved the best detection results in the original domain although some targets appear incomplete. OSP and SMF failed to effectively separate targets from the background. The proposed method, under different CRs, almost completely extracted the targets in CS but did not entirely suppress the background in this dataset.

In the case of the ABU-Airport dataset, ECEM exhibited the most favorable results in the original domain, while ACE and SMF detected some targets, and OSP and HTD-IRN could not effectively suppress the background. The proposed method's detection performance was optimal at CR = 0.75, approaching the results of ECEM in the original domain.

For the ABU-Beach dataset, both ECEM in the original domain and the proposed method at different CRs demonstrated ideal detection results. In comparison, ACE and SMF failed to detect complete targets, while OSP and HTD-IRN could not entirely suppress the background.

Regarding the ABU-Urban dataset, ECEM exhibited the best detection results in the original domain, with other algorithms failing to effectively suppress the background or detect complete targets. The proposed method demonstrated optimal background suppression at CR = 0.1 and 0.25, with the completeness of target detection almost comparable to ECEM in the original domain across different CRs.

The proposed method tends to get smoothed target contours in the detection results. This is due to the utilization

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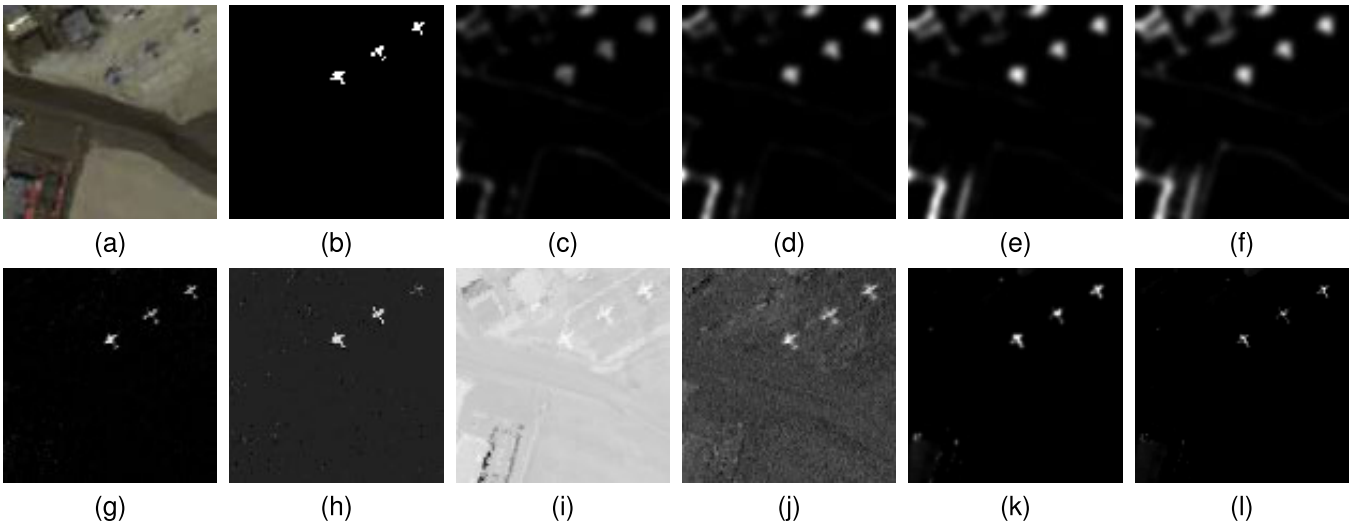


Fig. 6. Detection results of the Sandiego dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.1. (d) Ours CR = 0.25. (e) Ours CR = 0.5. (f) Ours CR = 0.75. (g) ACE. (h) ECEM. (i) OSP. (j) SMF. (k) HTD-IRN. (l) MLSN.

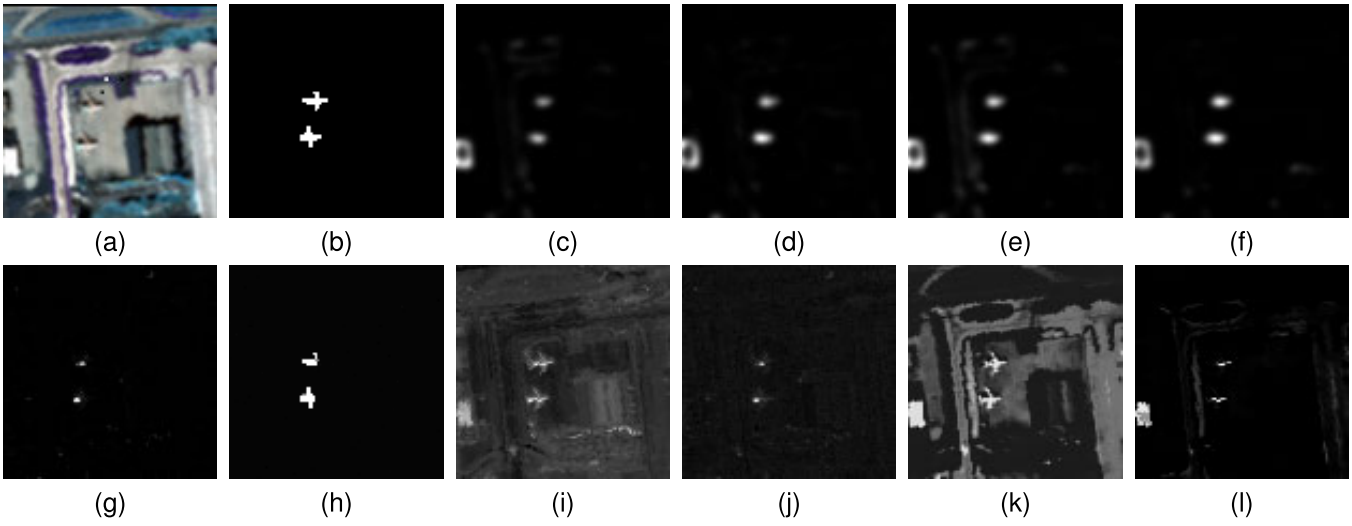


Fig. 7. Detection results of the ABU-Airport dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.1. (d) Ours CR = 0.25. (e) Ours CR = 0.5. (f) Ours CR = 0.75. (g) ACE. (h) ECEM. (i) OSP. (j) SMF. (k) HTD-IRN. (l) MLSN.

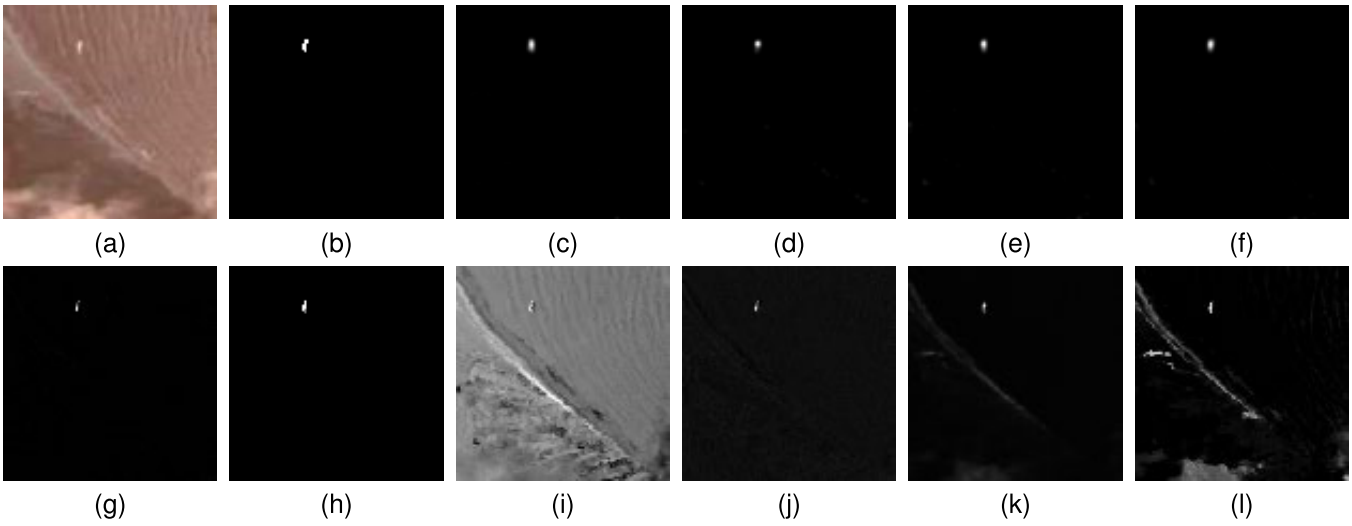


Fig. 8. Detection results of the ABU-Beach dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.1. (d) Ours CR = 0.25. (e) Ours CR = 0.5. (f) Ours CR = 0.75. (g) ACE. (h) ECEM. (i) OSP. (j) SMF. (k) HTD-IRN. (l) MLSN.

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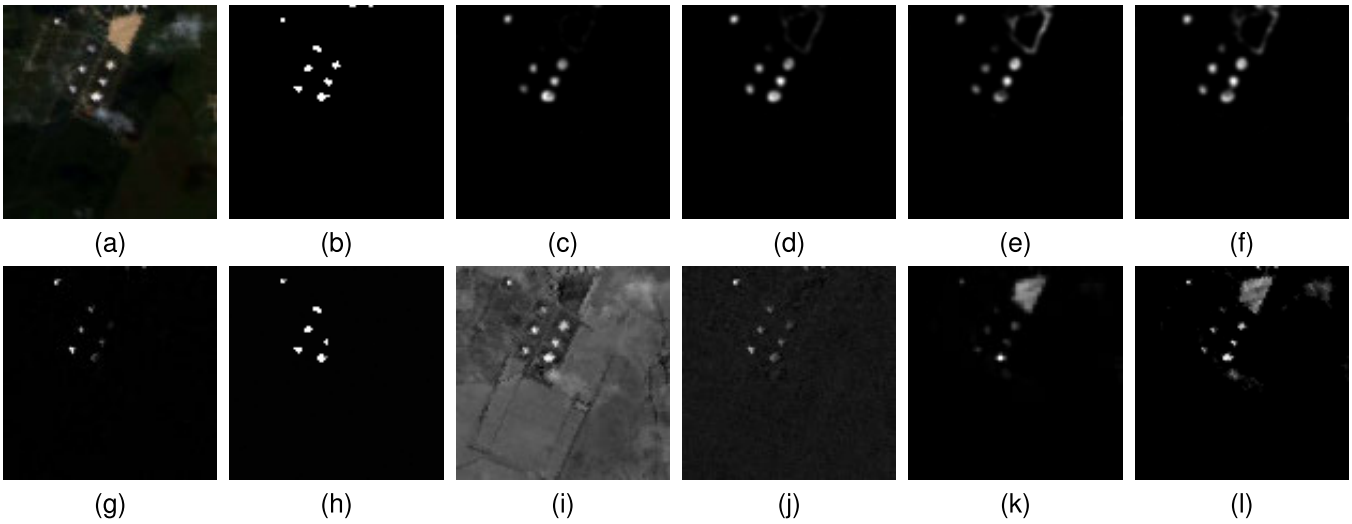


Fig. 9. Detection results of the ABU-Urban dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.1. (d) Ours CR = 0.25. (e) Ours CR = 0.5. (f) Ours CR = 0.75. (g) ACE. (h) ECEM. (i) OSP. (j) SMF. (k) HTD-IRN. (l) MLSN.

of neighborhood subspaces as sampling windows. Pixels near concave parts of target contours are enveloped by the target region in the sampling process, leading to misjudgments. In addition, convex parts of target contours may be misidentified as background.

3) ROC and AUC Analysis: The ROC curves for the Sandiego, ABU-Airport, ABU-Beach, and ABU-Urban datasets are displayed in Figs. 10–13. From left to right, the three curves are (P_d, P_f), (P_f, τ), and (P_d, τ). The overall performance of the proposed method is particularly outstanding in the ABU-Beach dataset, surpassing most algorithms in the original domain for ABU-Airport and ABU-Urban datasets, with slightly lower performance in the Sandiego dataset. The curves of the proposed method exhibit a sharp decrease in the false detection rate as τ increases, especially in the latter three datasets, demonstrating superior background suppression compared to most contrastive algorithms.

Tables II–V show the ACU values and running times for each comparative method across different datasets. The bold represents the best result while underlining the second.

AUC(P_d, P_f) reflects the overall performance of detection and false alarm rates. The proposed method achieves the maximum AUC(P_d, P_f) in the latter three datasets. ACE demonstrates the minimum AUC(P_f, τ), reaching about 1e-3 in all datasets. The proposed method closely follows ACE in this metric, achieving comparable background suppression performance in most detection results, maintaining AUC(P_f, τ) around 1e-2 in other scenarios. On the other hand, ECEM achieves high detection rates, while ACE performs poorly, and the proposed method ranks just below ECEM. In most situations, the proposed method can achieve the best or second all-around performance on AUC_{OA} and AUC_{SNPR},

4) Running Time: Considering that comparative methods are applied on the original domain, the running times in Tables II–V contain the reconstruction time and the HTD processing time. In terms of average HTD processing time for each comparative method, SMF is the fastest method in terms of average processing time, while ECEM has the slowest processing time due to the cascaded layers of CEM detectors, which sacrifices the processing time for performance. However, considering the reconstruction time, the proposed method greatly surpasses the comparative methods.

5) Ablation Study: In the ablation study, the AUC(P_d, P_f) based on different numbers of subsets N is compared, as shown in Table VI. The number of subsets is set as $N = [1, 2, 5, 10]$, and the AUC(P_d, P_f) generally increases with the increasing number of subsets. When the number of subsets N is 1, indicating no subset partition, the detection performance is the worst. Slight increases in the number of subsets lead to significant improvements in AUC. However, when the number increases to 5 and 10, further increasing N does not significantly improve performance but instead introduces greater computational complexity. Therefore, setting N to 5 is considered a relatively reasonable choice.

Fig. 14 illustrates the AUC(P_d, τ) of different CRs under whether band shuffle is used. It can be observed from the figure that introducing band shuffle can increase the AUC(P_d, τ) by about 0.05 in most situations, especially in a lower CR. This indicates band shuffle does have a positive impact on the detection rate.

F. Discussion

The experiments on public datasets, comparing our method with existing state-of-the-art ones in the original domain, demonstrate the excellence of our proposed HTD algorithm in CS.

1) From the analysis of RDP based on the QQ plot, when the sampling is from the single spectrum under CS, the fitting result with $N(0, B_L^2)$ distribution is well. Even when the sample number is rare with CR=0.1, all points are still distributed on the line. When the sampling is from the subspace within the neighborhood, some

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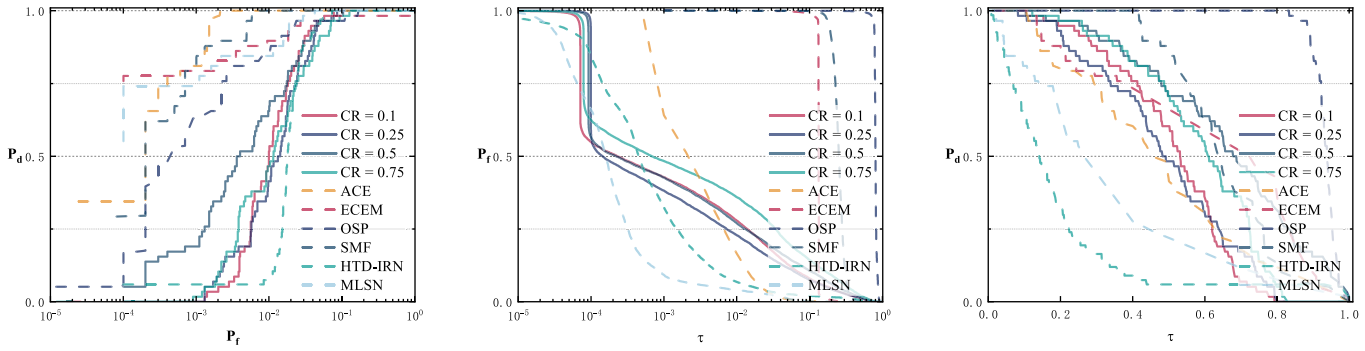


Fig. 10. ROC(P_d, P_f), (P_f, τ), (P_d, τ) of the Sandiego dataset.

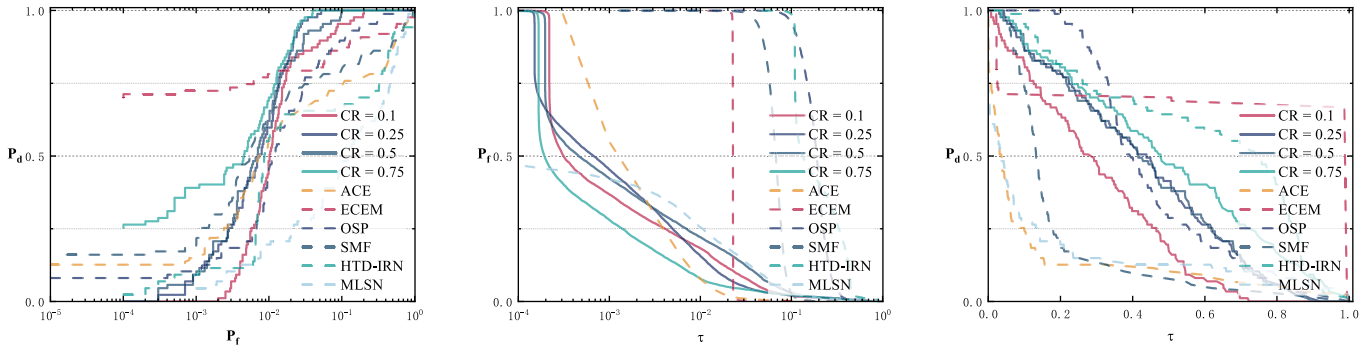


Fig. 11. ROC(P_d, P_f), (P_f, τ), (P_d, τ) of the ABU-Airport dataset.

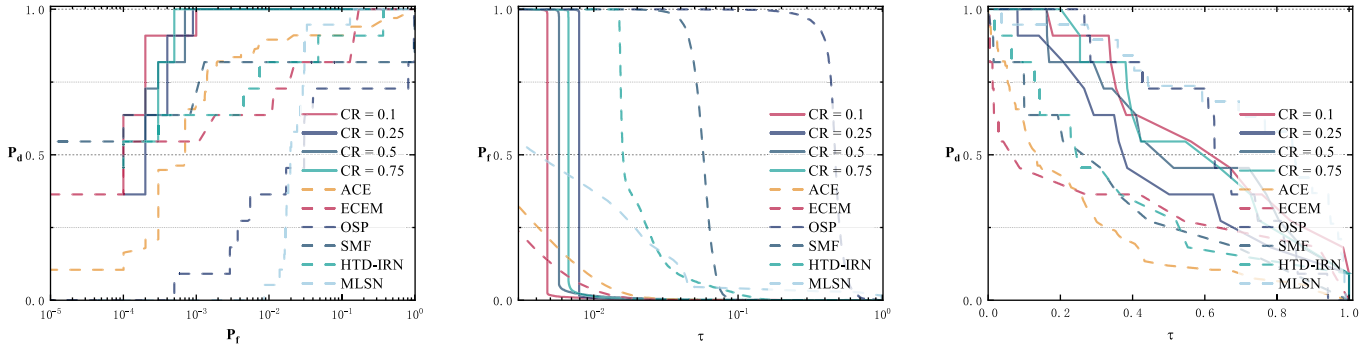


Fig. 12. ROC(P_d, P_f), (P_f, τ), (P_d, τ) of the ABU-Beach dataset.

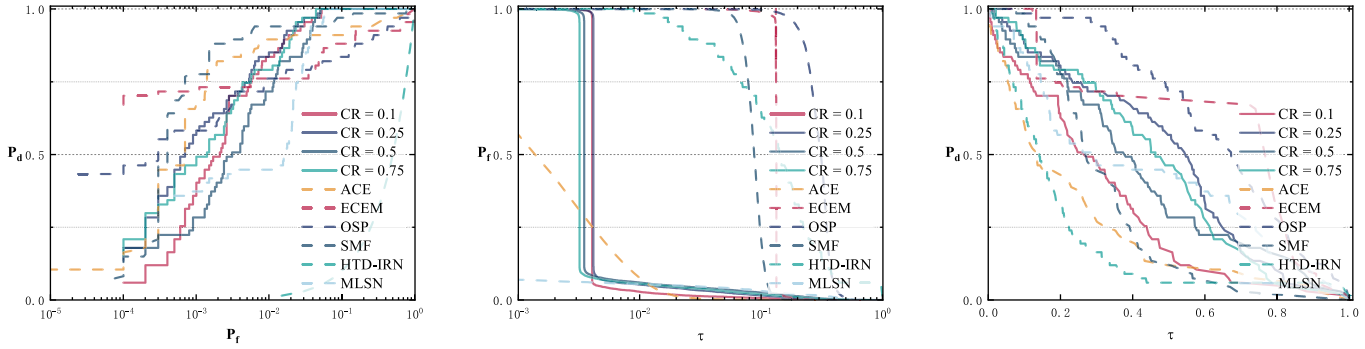


Fig. 13. ROC(P_d, P_f), (P_f, τ), (P_d, τ) of the ABU-Urban dataset.

deviation points appear at both ends of the line, and the deviation is aggravated when the sampling number decreases, but the points in the center are still closely arranged on the $N(0, B_L^2)$ line. This indicates that, in RDP, all vectors in the target subspace share the same Gaussian distribution in CS.

2) The proposed algorithm exhibits excellent adaptability to CR, demonstrating rapid convergence and maintaining

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TABLE II
QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR THE SANDIEGO DATASET

	$\mathbf{AUC}(P_d, P_f)$	$\mathbf{AUC}(P_f, \tau)$	$\mathbf{AUC}(P_d, \tau)$	$\mathbf{AUC_{OA}}$	$\mathbf{AUC_{SNPR}}$	**Times/s**
Ours CR=0.10	0.98532	0.03114	0.50788	1.46206	16.31125	**0.09503**
Ours CR=0.25	0.98229	0.02886	0.48773	1.44116	16.89916	0.12459
Ours CR=0.50	0.98751	0.03801	0.65539	**1.60490**	17.24258	0.15207
Ours CR=0.75	0.97999	0.05275	0.58241	1.50965	11.04093	0.19622
ACE	**0.99969**	0.00595	0.46977	1.46350	78.89525	0.09070+5.1348
ECEM	0.98033	0.13347	0.60938	1.45624	4.56567	4.31989+5.1348
OSP	0.99245	0.80660	**0.93431**	1.12016	1.15833	0.44243+5.1348
SMF	0.99916	0.25533	0.65121	1.39505	2.55051	**0.04224+5.1348**
HTD-IRN	0.99870	0.00360	0.53280	1.52790	148.00000	0.19696+5.1348
MLSN	0.99663	**0.00202**	0.33849	1.33310	167.83581	13.48604+5.1348

TABLE III
QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR THE ABU-AIRPORT DATASET

	$\mathbf{AUC}(P_d, P_f)$	$\mathbf{AUC}(P_f, \tau)$	$\mathbf{AUC}(P_d, \tau)$	$\mathbf{AUC_{OA}}$	$\mathbf{AUC_{SNPR}}$	**Times/s**
Ours CR=0.10	0.97762	0.01288	0.29888	1.26361	23.19715	**0.09301**
Ours CR=0.25	0.98985	0.01229	0.42252	1.40008	34.38297	0.10906
Ours CR=0.50	0.98659	0.01633	0.42219	1.39245	25.84860	0.15802
Ours CR=0.75	**0.99236**	0.00949	0.49056	1.47343	**51.70446**	0.17993
ACE	0.85075	**0.00371**	0.12636	0.97340	34.05023	0.09812+4.85192
ECEM	0.92659	0.02281	**0.69315**	**1.59693**	30.38799	5.77801+4.85192
OSP	0.95775	0.19355	0.45194	1.21613	2.33493	0.43730+4.85192
SMF	0.88864	0.06793	0.18673	1.00744	2.74893	0.04176+4.85192
HTD-IRN	0.83180	0.22130	0.59430	1.20480	2.68549	0.22093+4.85192
MLSN	0.75516	0.01773	0.16161	0.89904	9.11391	14.24632+4.85192

TABLE IV
QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR THE ABU-BEACH DATASET

	$\mathbf{AUC}(P_d, P_f)$	$\mathbf{AUC}(P_f, \tau)$	$\mathbf{AUC}(P_d, \tau)$	$\mathbf{AUC_{OA}}$	$\mathbf{AUC_{SNPR}}$	**Times/s**
Ours CR=0.10	0.99983	0.00521	0.63154	1.62616	121.29683	**0.08764**
Ours CR=0.25	0.99975	0.00827	0.47122	1.46271	57.00057	0.10678
Ours CR=0.50	**0.99980**	0.00631	0.56454	1.55802	89.43901	0.14069
Ours CR=0.75	0.99983	0.00715	0.58545	1.57813	81.87099	0.17318
ACE	0.97126	0.00236	0.31220	1.28111	132.45052	0.09870+4.95172
ECEM	0.98678	**0.00139**	**0.77232**	**1.75771**	**555.62590**	4.77721+4.95172
OSP	0.73641	0.47253	0.60458	0.86846	1.27945	0.48565+4.95172
SMF	0.81799	0.05747	0.33436	1.09488	5.81824	0.04213+4.95172
HTD-IRN	0.96110	0.02390	0.35670	1.29390	14.92469	0.21111+4.95172
MLSN	0.96225	0.03347	0.55489	1.48367	16.57835	16.83527+4.95172

TABLE V
QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR THE ABU-URBAN DATASET

	$\mathbf{AUC}(P_d, P_f)$	$\mathbf{AUC}(P_f, \tau)$	$\mathbf{AUC}(P_d, \tau)$	$\mathbf{AUC_{OA}}$	$\mathbf{AUC_{SNPR}}$	**Times/s**
Ours CR=0.10	0.99377	**0.00370**	0.30341	1.29349	**82.00281**	**0.08811**
Ours CR=0.25	**0.99489**	0.00899	0.48073	1.46663	53.45797	0.10788
Ours CR=0.50	0.99051	0.01080	0.41951	1.39922	38.85250	0.14999
Ours CR=0.75	0.99422	0.00953	0.45106	1.43576	47.34250	0.18923
ACE	0.96624	0.00419	0.23616	1.19820	56.37153	0.09367+5.08922
ECEM	0.92325	0.00907	**0.68225**	**1.59643**	75.22051	5.93833+5.08922
OSP	0.92347	0.30421	0.64105	1.26031	2.10727	0.36729+5.08922
SMF	0.98353	0.08867	0.31708	1.21194	3.57596	0.04279+5.08922
HTD-IRN	0.97930	0.00980	0.20020	1.16970	20.42857	0.22575+5.08922
MLSN	0.98425	0.01071	0.42929	1.40282	40.08379	17.04815+5.08922

high performance as CR increases. In datasets with complex backgrounds, such as ABU-Airport, only our method and ECEM can detect complete targets with minimal background interference. For datasets with small targets, such as ABU-Beach and ABU-Urban, our method maintains background suppression capability and target integrity across different CRs, surpassing many existing algorithms in the original domain.

3) The detection performance of our method on datasets where the distributions of targets and backgrounds are

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TABLE VI
COMPARISON OF AUC(P_d, P_f) WITH DIFFERENT SUBSET NUMBERS ON FOUR DATASETS

		Number of subsets			
		$N=1$	$N=2$	$N=5$	$N=10$
Sandiego	**CR=0.10**	0.95265	0.97460	0.98109	0.98363
	CR=0.25	0.97751	0.97973	**0.99021**	0.98859
	CR=0.50	0.97519	0.97786	0.98578	**0.98642**
	CR=0.75	0.97825	0.97969	**0.98640**	0.98152
ABU-Airport	**CR=0.10**	0.90122	0.95826	**0.97762**	0.97504
	CR=0.25	0.92407	0.96627	**0.98985**	0.98729
	CR=0.50	0.96881	0.96857	0.98659	**0.98736**
	CR=0.75	0.98019	0.99184	0.99236	**0.99392**
ABU-Beach	**CR=0.10**	0.72735	0.99956	0.99983	**0.99986**
	CR=0.25	0.90883	0.99963	0.99975	**0.99991**
	CR=0.50	0.99963	0.99971	0.99980	**0.99985**
	CR=0.75	0.99978	0.99975	0.99983	**0.99988**
ABU-Urban	**CR=0.10**	0.67927	0.91469	0.99377	**0.99620**
	CR=0.25	0.83339	0.91250	0.99489	**0.99747**
	CR=0.50	0.84116	0.95093	0.99051	**0.99614**
	CR=0.75	0.95184	0.96271	0.99422	**0.99755**

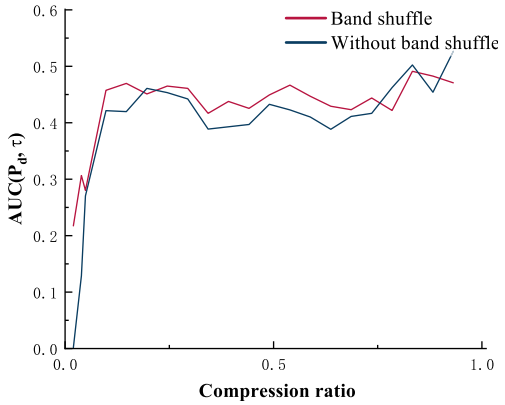


Fig. 14. AUC(P_d, τ) on ABU-Urban with different CRs under whether band shuffle is used.

similar still needs improvement. For example, in the Sandiego dataset, although complete targets are detected, the background is not suppressed enough compared to ECEM and ACE algorithms. During the CS process, RDP preserves the distribution characteristics of spectral random vectors. The distribution of targets in CS is related to the distribution of spectral vectors in the original domain. Therefore, objects with distributions similar to the target in the original domain may lead to false alarms.

4) Our method simultaneously meets the requirements for timeliness and high performance. The HTD latency is much lower than that of the ECEM, OPS, and HTD-IRN algorithms, despite some of them achieving ideal detection results. The advantage of the proposed method is to avoid the time consumption of reconstruction. Moreover, the proposed front-end HTD method abjures the impact of reconstruction performances on HTD results.

VI. CONCLUSION

This article mainly focuses on the problems of front-end HTD to solve traditional methods that cannot be directly applied to compressed hyperspectral without reconstruction.

First, a novel CS property called RDP is introduced, which is analogous to RIP, REP, and RSP.

Second, leveraging RDP, a detector based on spectral distribution estimation is developed. This detector uses KL divergence to estimate the similarity between the subspace sample and the target, achieving target detection under a compressed domain for the first time.

Finally, experiments indicate that the proposed method strikes a balance between background suppression and target detection capabilities, even outperforming some existing algorithms in the original domain. The introduced method also demonstrates low processing latency, meeting the timeliness requirements for dynamic front-end HTD.

Future works will delve further into HTD under compressed domain, exploring DL methods to replace similarity calculations based on KL divergence, aiming to enhance the capabilities of front-end HTD.

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